



ixia
A Keysight Business

The
ABCs
of Inline
Security

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Introduction

Strong security involves both inline and out-of-band monitoring, but this book focuses specifically on concepts related to solutions deployed directly in the flow of live network traffic to identify and prevent cyberattacks and other security incidents.

The definitions in this book provide a starting point to understand how to improve the efficiency and resilience of your inline security systems. While some of the terms can be intimidating, these definitions will hopefully make them easier to understand. A simple Definition, Use Case, Considerations format allows you to quickly understand what the concept is, where it is applied, and what points to consider when investing in this type of functionality.

Use the following diagram as a reference for understanding how a visibility architecture improves inline security monitoring and resilience. Note there are three basic steps to inline monitoring:

- The first step is **data access**, in which bypass switches intercept network traffic so it can be validated before it is forwarded into the trusted network.
- The second step is **data processing**, in which network packet brokers filter and pre-process packets, so only relevant data ends up being sent to the security solutions.
- The last step is **security monitoring**, in which the relevant data is inspected for specific threat signatures and suspicious traffic is removed before packets are forwarded on to their final destination.

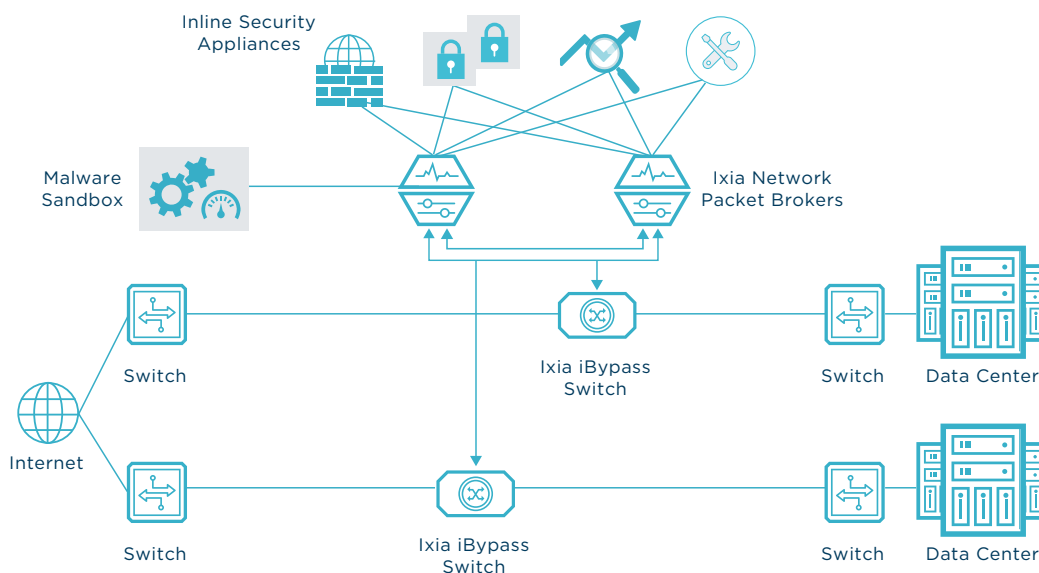


Figure 1. Visibility platform deployed to improve online security monitoring and resilience

Active vs Standby

Modern IT environments don't tolerate delay very well—even if it's just seconds. Customers, clients, employees, and partners demand instant response and fast performance. This has increased interest in having a secondary, redundant node move from being a standby to being an active participant in normal operations and reduce any delay in recovery.

PURPOSE OF ACTIVE AND STANDBY

High availability (HA) is achieved through the presence of redundant devices that can back each other up in the event that one of the items fails. The earliest implementations of high availability focused on having a second device in standby mode, sort of like an understudy waiting to step in should the principal dancer be unable to perform. While more than two systems can be involved, the most common implementation is still a primary and secondary node.

The Active/Standby mode of operation (A/S) evolved from initially having the hardware available and connected into the network, to having the necessary software completely installed and available on the second node as well. This is sometimes referred to as a 'warm standby' or 'warm spare.' The next level up is not only having the software installed, but also having the data stored in a way that it is immediately available to the secondary node as well. This is referred to as a 'hot standby.' The cost of a hot standby is a bit more, but the cost of downtime to an organization can be substantial, so most standby configurations today are of the 'hot' variety. Most hot standby implementations recover in some fraction of a minute or seconds.

One step up from a hot standby is the Active/Active (A/A) configuration. In this mode, both devices are actively processing data and share the workload although they are sized so that one device can completely take over processing in the event one fails. In this configuration, congestion is unlikely and failover happens instantaneously.

TYPICAL USE CASES AND BENEFITS

Creating resiliency is a part of every IT implementation. With the growth of cyber-attacks and the rising cost of data breaches, ensuring the resiliency of network security is more important than ever and a key use of A/A and A/S configurations.

Redundant network paths

Good network design involves creating redundant paths to maintain operations in the event of a path failure. Alternate paths can be configured to provide visibility to traffic in normal operations (active mode) or to begin operating only if the primary path stops or slows to an unacceptable rate (standby mode). All network paths, whether in A/S or A/A, must have visibility solutions attached, to ensure there are no blind spots where attacks could pass through unnoticed.

Resilient security architecture

Many security monitoring tools and appliances are configured in pairs to achieve HA and minimize downtime in the event of a hardware failure. Tool vendors generally require only a single license to operate an HA pair and the second node is deployed as a ‘warm standby’ that is activated only in the event the primary device goes offline. When that occurs, an alarm is activated to alert security personnel to the failover.

Security fabric with network packet brokers

Redundancy is also key in a security architecture design, where network packet brokers (NPBs) filter packets and perform preliminary processing of live traffic before passing to the security tools. NPBs keep security tools—many of which are major capital investments—working efficiently and perform critical tasks such as decryption and load balancing. Ixia is the only vendor that offers NPBs with the capability of being configured in A/A mode, using a dedicated HA link for complete synchronization. Both nodes are actively working and each node is aware of the traffic being processed on the other at all times. In the event of an NPB failure, recovery is instantaneous, with near-zero packet loss.

CONSIDERATIONS WHEN EVALUATING FAILOVER OPTIONS

When evaluating your options for failover in your network visibility and security architecture, consider the following:

Speed of Recovery

As mentioned earlier, if the speed of recovery in a failover situation is key, an A/A configuration provides the fastest possible recovery. Both NPBs are actively engaged in traffic processing and there is no time required to transfer processing to the standby node. This can be key to inline security processing, where tools are actively inspecting traffic in real-time. Other applications—such as out-of-band traffic analysis—can accommodate more latency in failover and may be served well enough by an A/S configuration.



Tolerance for Risk

An A/S configuration introduces some additional risk that the standby device has failed silently and will not be able to take over processing in the event the primary device fails. In highly-sensitive environments, this could be an important reason to operate in A/A mode.

Budget Restrictions

The cost of redundant architecture can be a substantial portion of the overall security budget. Many organizations have found it easier to justify the cost of a redundant NPB if it can be put to use immediately in normal operations as an active node. It is the duty of the security team, however, to make sure usage of the two NPBs combined do not exceed 50% of total, because one NPB must be able to completely handle all of the traffic, in order to provide complete failover.

Maintenance without Disruption

In an A/A configuration, maintenance can be completed without any special scheduling and without disruption to the network. Traffic is temporarily routed through the backup devices with no noticeable impact to users. In fact, periodic maintenance proves that the existing configuration is still able to handle the volume of traffic being processed. Any slowness experienced during a maintenance activity would indicate the need to scale the system.

Port Redundancy

The chance of a single port on an NPB failing is much greater than the entire device. Therefore, even in an A/S configuration, you may want to wire your inline tools to the Active NPB on redundant ports, so the secondary port can take over in the event the primary port fails, preventing a complete device failover. This is less important in an A/A configuration since the workload is shared.

Whatever failover mode works best for your environment, make it a priority. Lack of visibility in a complex system puts the entire infrastructure at risk. You want to be able to intervene quickly, accurately, and effectively to protect your network.

MORE INFORMATION

Find out more about resilient configurations:

- White paper: [Architecting for Security Resilience](#)
- Solution brief: [Creating Resilience Within Security Architecture](#)
- Solution brief: [High Availability Makes Monitoring More Reliable](#)

Blind Spots

Blind spots refer to portions of network traffic that are not being observed by your security and performance monitoring solutions. This can be due to an architectural inhibitor, such as virtual networking, or simply because you have not yet deployed a network tap to access traffic on a particular network segment. Blind spots can lead to network problems and outages, increased network security risk, and potential regulatory compliance issues.

EXAMPLES OF NETWORK BLIND SPOTS

While several of these may not apply to your organization, scan the list to familiarize yourself with the common causes of blind spots.

Organization silos

IT security, networking, and compliance groups may not frequently interact or share data and may form silos within an enterprise. This can lead to inconsistent data and compliance policies, port contention issues, improper programming that results in incorrect or missing data captures, and data conflicts that arise from collecting data at the wrong places.

Virtualization technology

According to Gartner Research, up to 80% of virtualized data center traffic is east-west (i.e. inter- and intra-virtual machine traffic) so it never flows to the top of the rack where it can be monitored by traditional tap and SPAN technology. While virtual tap technology exists to counter this threat, IT staff may not yet have implemented the technology.

Rogue IT

When users add their own Ethernet switches and access devices, use cloud-based data storage, or add something else to the network, company security policies are often subverted. This can open the door to security, compliance and liability issues. IT is rarely in control of these resources, especially as they can appear sporadically, like Wi-Fi hot spots.

Mergers and acquisitions

The blending of disparate equipment and systems often causes interoperability issues which can lead to system/application downtime, system capabilities being turned off to improve network performance, and the scaling back/elimination of network and application monitoring. This can result in limited visibility or blind spots where no one really knows what is happening.

Addition of new network equipment

When new equipment is added, there is often no record as to who owns it and what it does. The equipment can be forgotten about, especially when IT personnel leave the company or change departments. “Lost” equipment that is still functioning in the network can be a source of security vulnerabilities due to lack of proper software updates and unknown user access privileges.

Inconsistent monitoring/data collection policies

This can occur in multiple situations, but one common reason is that virtual monitoring and physical monitoring policies differ, which can cause compliance issues when data is simply not captured.

Network upgrades are postponed

Postponing equipment upgrades can cause network performance to slow, which affects IT’s ability to solve problems as fast as required. More information is available in this [e-book](#) on optimizing network visibility.

Network upgrades are implemented

Even making necessary upgrades can result in blind spots. One example is when new, higher-speed equipment is added. This equipment may end up overloading other components of the network, especially monitoring and security tools, with too much data. This is especially true if those tools weren’t upgraded at the same time. Overloaded tools can lose (i.e. drop) data or overwrite buffers/logs at a faster than expected pace.

Security audits are postponed or overdue

Lack of security oversight and corrections can result in a safe and cozy harbor for various attacks and malware on your network. It’s hard to say what will be hidden, but whatever it is, you probably don’t want it. See this [resource](#) for more information.

Configuration mistakes

An example of this is firewall programming, which is rules-based and typically processed through comparison to access lists. When the traffic matches a rule, it is immediately forwarded on, even if subsequent rules exist to tailor the information. This can cause gaps in network security because the packet was routed before the correct security tool got to see the correct information.



CONSIDERATIONS FOR ELIMINATING BLIND SPOTS

With the blind spots listed above, you typically have two ways to respond—either in a proactive or reactive manner. The reactive approach is straight forward: just wait until an issue arises and then fix it. While it's the simplest approach, it is also usually the costliest in terms of the time it takes to locate the blind spot causing the issue. In addition, it often necessitates the purchase or implementation of capital equipment or multiple “Band-Aids” that never really solve the problem.

If you want to follow a proactive approach, the best solution is to design a visibility architecture. (See entry on page xx). This involves some upfront cost and planning but will normally pay for itself very quickly. Your visibility architecture is a plan for providing network data access to your security and monitoring tools. This involves how data is accessed (taps or SPAN ports), where tools are connected (edge, core, which branches, etc.), and how you choose to process the data before you send it to a tool (packet filtering, application filtering, deduplication, packet trimming, decryption, aggregation, etc.).

Unknown issues and “soon to be problems” exist in every network to some degree. To achieve the goal of ending blind spots in your network, you need see everything in your network and implement a total visibility architecture. It's not hard or complicated and the sooner you do it, the sooner you can strengthen your inline security, reduce operating costs, and improve IT productivity.

MORE INFORMATION

Find out more about eliminating blind spots in your network:

- E-Book: [Network Visibility For Dummies](#)
- White paper: [Visibility Architecture—Eliminating Blind Spots](#)
- White paper: [Five Steps to Building Visibility and Security into Your Network](#)

Bypass Switches

The latest inline security tools can prevent network downtime by keeping threats at bay. Yet they come with risks too. Inline tools are single points of failure in your network, meaning that if they fail, the whole network fails with them. Bypass switches can save you from this unplanned outage.

WHAT IS A BYPASS SWITCH?

A bypass switch allows you to connect inline security tools to your network—without the risk of network downtime. When an inline tool fails, bypass switches automatically kick in. They redirect network traffic around the tool so it flows around the failed tool, instead of through it. The network traffic is able to “pass by” the blockage caused by the tool. So the network stays up and running—even if the tool doesn’t.

Bypass switches detect when an inline tool has failed or lost power through the use of heartbeat packets. Heartbeat packets are signals sent from the bypass switch to the inline tool at regular intervals. If the tool does not return the packet to the bypass switch at the interval expected, the inline tool is assumed to have failed and network traffic is rerouted.

There are two basic types of bypass switches: internal and external.

Internal bypasses

Some firewall and intrusion protection system (IPS) appliances are designed with an internal bypass embedded inside. The vendor is therefore able to claim the network is fully protected in the event their device fails for any reason. Most people assume this is adequate protection, but an internal bypass may not actually be the best solution. What if the appliance doesn’t fail completely and, instead, only slows response times? In that situation, the external bypass offers a better solution.

External bypasses

An external bypass switch is a simple piece of hardware designed to be deployed on the live network in front of a security monitoring appliance. Compared to the internal bypass embedded in a sophisticated security tool, the mean time between failure of an external bypass is orders of magnitude longer. Other advantages of external bypass switches are described below.

USE CASES AND BENEFITS

Increase network reliability

The bypass switch is a passive device that is installed directly on the production network in front of any appliance that processes live network traffic (like an IPS or NGFW). The bypass continually monitors inline security solutions at very fast intervals and reacts quickly to reroute traffic if it fails to receive the expected heartbeat. The faster the bypass reacts, the less risk there is that transactions or interactions will be affected and network availability is maintained. The cost of network outages is well documented and can include a loss of revenue and an impact to brand and reputation. A bypass easily pays for itself by reducing the risk of a network outage.

Maintenance and upgrades without downtime

An external bypass also allows you to proactively take an inline security appliance offline for maintenance or troubleshooting without affecting network availability by manually putting the switch in 'Bypass On' mode. Then, with live traffic no longer flowing through your security solution, you can troubleshoot or even replace the device without impacting network availability.

Increase security resilience

Another compelling benefit of the external bypass is that it has both a primary and secondary connection to help you maintain security monitoring during a failure of the primary device. With an internal bypass, when a device goes down, traffic is routed around it while troubleshooting takes place. The network remains available, but the security inspection provided by that device temporarily ceases until the device comes back online. With a product like the [Ixia iBypass](#), traffic can be automatically sent to a backup device until the primary device is back online, with no manual intervention required. Your security architecture is more resilient. This capability can be critical in environments like financial services or healthcare. With a primary and secondary path, you can keep security inspection strong and the network available, even during failure of a key security appliance.



CONSIDERATIONS WHEN EVALUATING BYPASS SWITCHES

When evaluating bypass solutions for inline security, consider the following:

External vs Internal Bypass

At some point, your inline security tool will need to be taken offline and physically removed from the network. This may be for planned maintenance or upgrade purposes. But it could also be due to an unplanned event. If the tool has been installed behind an external bypass switch, it can be removed without impacting the flow of network traffic.

The image below shows how an **Intrusion Prevention System (IPS)** can be disconnected without breaking any network links.

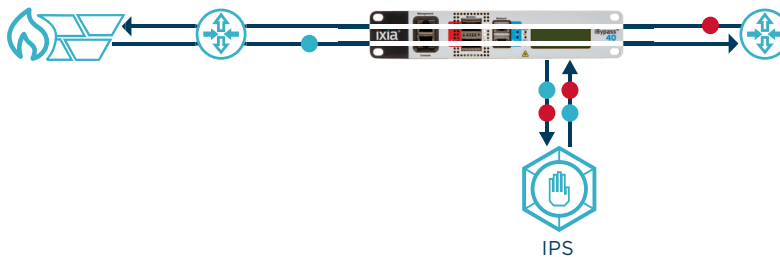


Figure 2. External bypass switch

In contrast, removing an IPS with an internal bypass switch disrupts the network link, which creates costly downtime and service interruption for your organization.

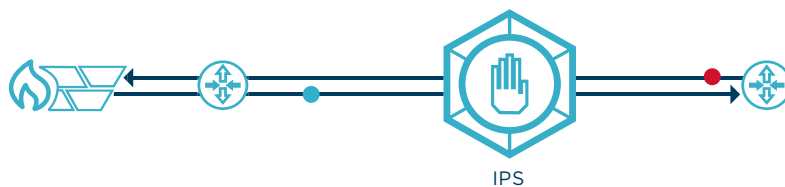


Figure 3. Internal bypass switch

Cost of Backup

The purpose of the bypass is to prevent a network failure, but what happens if the bypass fails? If you do not have a backup device available, a network outage will result. Knowing the probability of the bypass failing can help you determine an appropriate backup strategy. Physical devices generally come with an estimate of their useful life, expressed as the mean-time-between-failure (MTBF). Although an internal bypass is convenient, it is embedded within a much more

complex device with many moving parts and a much lower MTBF. In contrast, an external bypass switch is a very simple device with an MTBF approximately five times greater. Given the low risk of failure and the reasonable cost of an external bypass, you can typically configure a backup strategy at much less cost than relying on internal bypass functionality.

Negative Heartbeats

Bypasses monitor the status of security appliances using very small heartbeat packets that are continually sent from the bypass at very fast regular intervals. Some bypasses continue to send heartbeat packets even after a device goes offline, referred to as sending “negative heartbeats”. This allows the bypass to quickly determine when a device is back online and shift back to sending traffic to the now-functioning device.

Remote Bypass Management

Another useful feature of the iBypass is the remote management interface that lets you monitor the switch, as well as the status of the attached appliances and links. The interface also allows you to make changes if you need to, without having a technician onsite. This feature is particularly useful for ensuring your security devices are updated with all of the latest software updates and releases. Cyber criminals have a lot of success taking advantage of known vulnerabilities. They know security teams are overworked and maintenance windows are fully booked for weeks. Even when an exploit is published and fixes are available, the window to take advantage of the hole can remain open for months or even years. Being able to easily suspend the processing of the device and install the update from a remote location makes it that much easier to keep your security tools as strong as possible.

The management interface also supplies useful traffic statistics. Bandwidth utilization, peak traffic, packet and byte counts, and number of errors help you perform basic network monitoring without investing in additional tools. You can also configure the bypass to generate alarms when traffic exceeds a chosen level and trigger a response from another network management tool.

MORE INFORMATION

Find out more about the value of bypass switches:

- Solution brief: [Improve Network Reliability with External Bypass Switches](#)
- Solution brief: [Strengthen Security and Reduce Risk of Downtime](#)
- [Bypass FAQs](#)



Fail Closed

Confirm your understanding of similar sounding terms related to maintaining network availability during inline security monitoring.

EXPLANATION OF FAIL CLOSED

Simply stated, failing closed is when a device or system is set, either physically or via software, to shut down and prevent further operation when failure conditions are detected. This strategy is common in situations where security concerns override the need for access. We encounter this every day when we forget the password to a seldom-used personal account and are denied entry. A physical example is the failure of a metal detector at the entrance to a federal courthouse, which leads to a long line of people waiting to get in at a second door, while a technician tries to repair the first door. In these situations, access is a second priority to security.

USE CASE & BENEFITS

The primary use case for Failing closed is to prioritize network security. In an IP network, security appliances like firewalls can be configured to fail closed which prevents incoming Internet traffic from being passed into your internal network when the firewall is unable to confirm that the packet is allowed. The network outage that results from a firewall outage can be minimal, if a backup firewall quickly takes over processing duties (like the second door at the courthouse). The fail closed condition generally provides greater confidence that a cyber threat or attack will not sneak in while a firewall is offline.

CONSIDERATIONS

It's important to note that the fail closed strategy, even for a device like a firewall, has not always been the rule. In some environments, network interruption can be a greater concern than security, leading to the choice to fail open. This was more frequently the case in the early days of firewall deployment, when organizations were learning how to balance the need for security inspection with network availability.

MORE INFORMATION

Find out more about the function of bypass switches:

- White paper: [External Bypass Switches: A Better Inline Security Tool](#)
- Buying Guide: [Which iBypass Switch Belongs in Your Network](#)

Fail Open

EXPLANATION OF FAIL OPEN

A system set to fail open does not shut down when failure conditions are present. Instead, the system remains “open” and operations continue as if the system were not even in place.

This strategy is used when access is deemed more important than authentication. Healthcare systems are sometimes operated on a fail open basis, such as when emergency care is provided even without authentication of insurance coverage or the ability to pay. The risk (of non-payment in this case) is essentially mitigated by performing authentication after-the-fact.

Another example often cited is when a door with an electronic locking mechanism is automatically unlocked when the system fails and is unable to authenticate access credentials. This ensures an exit is made available, particularly in the event of a fire or natural disaster that disables electronic systems.

USE CASE & BENEFITS

Protect network availability

Historically, some organizations considered inline deployment of a network firewall to be a “nice-to-have,” rather than an essential element of IT security. When a firewall failed, they preferred to have it fail open and let Internet traffic proceed on into the internal network without authentication. The thinking was that, the majority of traffic was safe and the risk of a network breach was low, so it did not make good business sense to interrupt network operations. The business risk was minimized by prioritizing firewall restoration to limit potential exposure and by analyzing copies of network traffic (using out-of-band tools) to detect suspicious activity after the fact. The fail open condition prevailed in situations where access was deemed more important than security.

Supplement other security appliances

There are security solutions that organizations may want to operate in a fail open condition to supplement the function of existing security appliances. One example is an advanced malware protection (AMP) sandbox, which is used to execute unknown files in a safe environment and provide the results to anti-malware solutions. Since the sandbox is supplementing the main device, its failure may not require a complete shutdown of processing.

Deployment and testing

Another practical use for fail open is during the initial deployment and testing period of a new security appliance. Configuring a new device to fail open allows the team to become comfortable with the operation and learn how to respond to alert situations without becoming overwhelmed. Once the team feels confident, the device can be switched over to a fail closed condition, for greater risk management.

CONSIDERATIONS

The primary consideration for a fail open situation really comes down to what is the design of your security architecture? This architecture will have a lot of impact on your choice.

MORE INFORMATION

Find out more about the function of bypass switches:

- White paper: [External Bypass Switches: A Better Inline Security Tool](#)
- Buying Guide: [Which iBypass Switch Belongs in Your Network](#)

Fail Safe

EXPLANATION OF FAIL SAFE

Fail safe refers to a device that is configured to protect all other components in the system from failure, in the event the device itself fails. Practically, this can have the same result as failing open, but fail safe is often achieved through addition of a separate device, known as a bypass switch.

USE CASE & BENEFITS

Bypass switches are deployed “in front of” network devices and work by establishing a direct connection to the device and monitoring its ability to receive and process traffic. This is achieved by sending a very small network packet, called a heartbeat packet, to the device at very fast intervals—generally one every couple microseconds. If the packet is returned, the bypass remains open; if the packet is not returned, traffic is bypassed around the device and moved along to the next switch in the network.

CONSIDERATIONS

Many network security appliances, such as next generation firewalls and IPS solutions, now include an internal bypass function. However, internal bypasses do not provide all of the functionality of an external bypass switch.

An external bypass switch deployed in front of a network device can be activated proactively by the IT staff, to take a device offline for regular maintenance, periodic troubleshooting, or repositioning in the network. The external bypass essentially removes a particular device temporarily from the active network, eliminating the need to wait for a network maintenance window to perform upgrades or respond to support issues.

MORE INFORMATION

Find out more about protecting inline security tools:

- Video: [Bypass Switches](#)
- White paper: [Evaluating Inline Security: Key Considerations](#)

Failover

EXPLANATION OF FAILOVER

Failover is the ability to recover the functionality of network devices that fail. This is a broader concept than fail safe, which only specifies only no adverse impact to other components. Failover implies recovery of functionality, achieved through redundancy.

USE CASE & BENEFITS

This capability is often integrated in to core pieces of equipment, e.g. network switches and packet brokers, to ensure maximum network uptime. While the capability can be including into security and monitoring tools themselves, many failure scenarios make the practicality of that solution null and void. This functionality is better off being part of a purpose-built solution.

CONSIDERATIONS

External bypass switches and network packet brokers are now available with the ability to designate an alternative path for traffic in the event of a network device failure. For example, should the primary IPS appliance fail, when the external bypass switch or network packet broker detects the failure (within microseconds of the event), the switch can automatically begin sending traffic to a secondary, backup appliance. This can be a cost-effective solution for achieving resiliency.

MORE INFORMATION

Find out more about achieving fast failover:

- Video: Network Visibility: [Resilience Matters](#)
- Partner brief: [Best Practice Deployment Guide: Imperva & Ixia](#)

Firewall

The term firewall comes from the construction industry where it describes a fire-resistant barrier intended to prevent or slow the spread of fire through a structure. In the digital world, a firewall is a security system that monitors and controls incoming and outgoing traffic based predominately on security rules.

FUNCTION OF A FIREWALL

A firewall can consist of software, hardware, or both and is used to filter data communications entering and leaving a private network or computer system. If a packet of information does not satisfy the parameters entered into the firewall, the packet can be stopped before it crosses the threshold or sent to another device for further analysis. Firewalls are customizable, which means you can add or remove filters as conditions change, though most commercially-available firewalls come with already established settings. For every type of firewall, there is a risk that vulnerabilities in their particular configurations, filters, or processes may allow an attacker to gain control over the firewall and get access to the network or host. This underlines the importance of keeping all firewall software updated.

Traditional firewall features include¹:

- Basic packet filtering
- Stateful multi-layer inspection
- Support for Network Address Translation (NAT)
- Termination of a virtual private network (VPN)

KEY TYPES OF FIREWALLS AND ADVANTAGES

Host-based

A host-based firewall is installed on each individual server and controls traffic into and out of that machine. This function may be available inside the operating system, such as with Microsoft Windows and Linux. Host-based firewalls provide backup protection in the event perimeter security fails, protection against threats from inside the organization or mobile threats like viruses and worms, and can also mitigate the risk of poorly-configured software. A host-based firewall can also be configured to support a single type of application and block everything else, for a more specific defense that is less likely to be exploited.

¹ NSS Labs: "Test Methodology for Next Generation Firewalls," 2016, [accessed online](#).

Host-based firewalls are frequently combined with network-based protection so even if a determined hacker is able to breach the perimeter security, they still have to deal with the security around each host.²

Next generation firewalls

(See description on page [43](#)).

Perimeter or network-based

A perimeter or network-based firewall establishes a barrier between a trusted internal network and an untrusted outside network, such as the Internet. This type of firewall compares each packet received to a set of established criteria to determine if a packet will be forwarded. Packets that do not satisfy the filter criteria are either dropped or sent to another area outside the network for further analysis. Filtering can be done based on different attributes of the traffic.

Packet layer filtering: Network layer firewalls, also called packet filters, operate at the network layer of the TCP/IP protocol stack (layer 3) and prevent packets from passing through the firewall unless they match packet layer attributes such as source IP address, source port, destination IP address or port, or protocols. The rules are either provided as defaults within the firewall appliance or are programmed in by the firewall administrator. Stateful firewalls keep track of all packets related to the same connection and consequently are able to process packets much more quickly.

Application layer filtering: These firewalls operate at layer 7 of the OSI stack and look at what a protocol is being used for. For instance, an application-level filter can tell the difference between HTTP traffic used to access a Web page and HTTP traffic used for file-sharing. While firewalls that filter at the application layer can provide considerable data security, they can sometimes significantly affect network performance.

Web application firewall (WAF)

WAFs are deployed in front of web servers to protect them against attacks, to monitor and control access, and to collect access logs for compliance and audit purposes. WAFs are traditionally deployed inline as a reverse proxy, meaning they retrieve resources on behalf of users from one or more Web servers. This allows them to load balance requests between servers and optimize content by compression and caching. Some WAFs can also be positioned out-of-band to work on a copy of network traffic. Cloud-based WAF, delivered as a service, has also become a valid option for some enterprises.

² Avishai Wool, "Business-Driven Security Policy Management" November 30, 2015. Accessed on Information-Age.com on October 25, 2017.

SPECIFIC USE CASES

Protection from brute force, DOS, and DDOS attacks

Firewalls use a variety of detection methods to prevent large-scale denial of service attacks that can make a website inaccessible to users or disable the security in order to gain administrative access. The real target of the attack can be to steal personal data, install a backdoor for a subsequent attack, or install malicious software to harness computing power for a botnet.

Prevent and monitor insider threats

Organizations generally publish policies for acceptable use of Internet resources by employees and other insiders. Firewalls can be used to enforce limits and prevent unsafe behavior, monitor usage, and document activity for auditing purposes. They can also be used to automatically discover all privileged users, verify identity before access to high-value assets, and track changes in behavior over time.

Secure application access

Besides network perimeter and end point security, firewalls are increasingly used to monitor and control access to specific applications. One of the important points to monitor in sensitive applications is that the application is encrypting personally identifiable information (PII) and maintaining compliance logs. Firewalls play an increasingly significant role in maintaining the availability and uptime of applications.³

Proactively support compliance

Firewalls can perform traffic logging and generate documentation to support compliance reporting for a wide variety of compliance regulations.

Improve incident response and forensics efficiency

Firewalls can be integrated with other security solutions to accelerate the response to security alerts and help staff investigate security incidents. Armed with access to all of the data flowing to and from web applications, a WAF, in particular, can observe vulnerabilities, information leaks, configurations errors and similar problems before an attempt is made to exploit them.

³ Tufin Technologies survey; <https://www.tufin.com/pr/tufin-survey-reveals-the-number-one-use-case-for-firewall-management-is-to-ensure-secure-application-connectivity/>

CONSIDERATIONS WHEN DEPLOYING FIREWALLS

Capacity for Customization

Security policies will need to be adjusted as threats evolve and the enterprise should be able to easily update firewall configurations and blacklists/whitelists as the result of security analysis and forensics work. Having a centralized management interface is important when updating a large number of firewall devices in a distributed organization. Some enterprises offload their firewalls by deploying a gateway appliance in front of the firewall to pre-filter out millions of suspicious IP addresses provided as part of a real-time security and threat intelligence feed and supplied by dedicated expert organizations. (See the entry for Threat Intelligence Gateway for more information.)

Maintenance without Disruption

Keeping a firewall updated with the latest hardware and/or software releases is critical to its defensive capability. If lack of personnel or time means your firewall is out-of-date, you may not be protected at all. Installing a bypass switch in front of your firewall can make updates and migrations much easier and eliminate downtime. A bypass can be set to forward traffic through an alternate device or just pass it along during a maintenance event so the security team does not have to wait for the next maintenance window.

Ability to Inspect Encrypted Traffic

With security attacks increasingly embedded in encrypted traffic, enterprises need to decrypt traffic to perform a full inspection. Some firewalls are capable of decryption, but are not able to share the resulting plain text traffic with other security solutions that do not have this capability. An alternative is to deploy a network packet broker with decryption capability in front of your other security solutions. The NPB decrypts the traffic one time and forwards the packets as you desire to other security solutions and then re-encrypts the traffic before forwarding on to its final destination. This can be a more efficient way of handling the process-intensive function of traffic decryption and re-encryption.

High Availability

This is generally achieved by deploying two identical firewalls with the same operating system on the same network segment. A heartbeat connection between the firewalls ensures seamless failover in the event one device goes down. An Active-Passive connection is the default, where one device remains in standby mode until the first device fails. Devices with dedicated HA ports





can provide direct communication between the management planes on the firewalls and enable an Active-Active setup where both firewalls are processing production traffic. In situations where having a completely redundant firewall appliance on a particular network segment is not economically feasible, redundant network packet brokers can be configured to perform the same cut-over in Active-Active mode. The advantage is that the NPBs are much less expensive than the typical firewall.

MORE INFORMATION

Find out more about how to support the function of network firewalls:

- Solution brief: [Get More Value from Security and Monitoring Tools](#)
- Solution brief: [Improve Network Reliability with External Bypass Switches](#)
- Webinar: [Resilient Security Monitoring: One Step at a Time](#)

Heartbeat

In technology, a heartbeat is a repeating signal generated by a piece of hardware or software used to indicate it is alive and operating normally. Typically, a heartbeat is sent at regular intervals of two or more per second. If the endpoint does not receive a heartbeat signal for more than a couple heartbeat intervals, the machine is assumed to have failed and corrective action is triggered.

FUNCTION OF A NETWORK HEARTBEAT

In network security, a very small heartbeat packet can be used to proactively detect when a security appliance is able to receive data traffic. The heartbeat is initiated from a device that listens for the return of the heartbeat from the monitoring device. This process proactively identifies when a monitoring solution is not available so traffic flow can be switched to another route to avoid an outage. This is critical during inline monitoring when live network traffic is being monitored. The most important characteristic of heartbeats is the frequency at which they are sent since this will determine how fast a failover or bypass can be executed. Another important factor is that heartbeats be sent from a device with a very long mean-time-between.

USE CASES AND BENEFITS

Protection of network availability

Inline tools are deployed directly in the flow of live network traffic in order to prevent malicious traffic from impacting an enterprise. If an inline security appliance stops working and is unable to forward packets on, the network can suffer an unexpected outage and disrupt services. Heartbeat packets ensure that live traffic does not stop flowing while an outage in a security device is resolved. This is important to protecting against business disruption and/or impact on quality-of-service.

Total security resilience

In environments where there is zero tolerance for an outage of the security system, heartbeats ensure that the system is configured to failover as fast as possible. With sub-second heartbeats, a bypass switch can route traffic through an alternate network segment or a backup security appliance nearly instantaneously to ensure traffic is always inspected.

Maintenance and automatic recovery

Bypasses can help minimize downtime during unplanned outages of your security tool or planned maintenance events. If a security tool is protected by an external bypass switch, traffic will be routed around the tool anytime it is offline

and the switch will automatically begin sending traffic through the tool once it goes back online. In cases where it is cost-prohibitive to have a backup device, a negative heartbeat can minimize the amount of traffic that is bypassed around an offline device.

Faster, more accurate solution deployments

When administrators install a new security solution behind a preconfigured bypass switch, traffic will start flowing immediately to that device, which simplifies the deployment and eliminates the chance of configuration errors delaying the security benefits.

CONSIDERATIONS FOR EVALUTING HEARTBEAT TECHNOLOGY

Heartbeat Frequency

While heartbeats are common to many bypass technologies, Ixia's heartbeat technology allows administrators to specify a very high frequency for heartbeat packets—down to one nanosecond. This guarantees that if an inline security device fails, the bypass will know nearly instantaneously and react.

Preconfigured Heartbeats

Heartbeat packets must be tailored to work specifically with each monitoring device to ensure the heartbeat can pass through the appliance without being blocked. Manually configuring a heartbeat is subject to errors that can take time to resolve. Ixia addresses this issue by providing preset heartbeat values for most available security monitoring solutions. The administrator simply selects the monitoring solution from a drop-down menu to properly configure the heartbeat, eliminating the time and risk related to manual configuration.

Negative Heartbeats

Some bypass switches continue sending and listening for heartbeat packets when a tool is inactive in order to know when the tool comes back online. Negative heartbeats (heartbeats that do not come back) can establish a self-healing system in which normal processing is automatically restored once an inactive device comes back online.

MORE INFORMATION

Find out more about the use of heartbeats in Ixia iBypass switches:

- Blog: [Heartbeats Stop Heart Attacks for Inline Security](#)
- White paper: [External Bypass Switches: A Better Inline Security Tool](#)



High Availability or Security Resilience

The emphasis in security has traditionally been on increasing the sophistication and intelligence of security appliances and solutions to identify and stop threats from entering the network. While this is clearly important, there is an even more critical need to keep the security systems operating and available, even in the event of a component failure. The ability of your security architecture to survive an outage, reroute live traffic as necessary, and maintain a strong defense is referred to security resilience.

PURPOSE OF A SECURITY FABRIC

The most straight-forward way to create resilience is to have a redundant standby device available for each component in your security architecture. This approach, however, is cost-prohibitive for most organizations and unnecessary when you create a “security fabric” to enable effective resource sharing among network segments and protect the network from disruptions, configuration errors, or the unexpected loss of power. Rather than placing security appliances directly inline on the live network, you can deploy them behind an external bypass switch and a network packet broker that give you greater control over live traffic inspection (see Figure below). Heartbeat technology lets you route traffic around any appliance that stops responding to ensure network availability and maintain service levels. The NPB filters traffic according to rules you specify, to deliver only relevant traffic to your security solutions and minimize congestion. Once established, a security fabric can serve both inline inspection of live network traffic, as well as out-of-band analysis used to identify multi-stage cyberattacks that are executed in phases to avoid detection by inline tools.

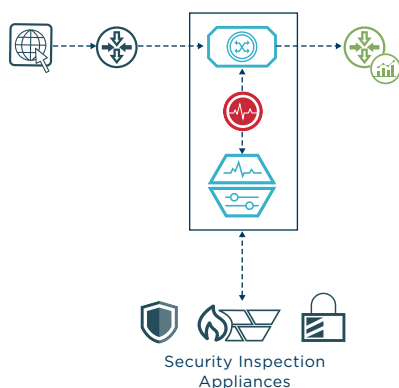


Figure 4. Security fabric includes bypass switch and NPB

OPTIONS FOR INCREASING SECURITY RESILIENCE

N + 1 deployment

Without a security fabric in place, each inline security solution deployed—no matter how much capacity is being used—would need a redundant standby to take over if the solution becomes unavailable due to an overloaded port, power failure, software configuration error, or the need for maintenance or troubleshooting. With a security fabric, tool capacity is shared across the network, which enables redundancy to be achieved with an N+1 tool strategy where N is the number of tools required for full capacity plus one additional, standby tool to cover any single device that stops responding. Tool sharing saves money by allowing multiple network segments to have access to the standby solution.

High availability security fabric

To keep traffic flowing in the rare event your bypass switch or NPB stops functioning, you can configure your security fabric for high availability, just as you do with your security solutions. With a redundant bypass switch and NPB, you can create dual paths to each of your security solutions and eliminate any single point of failure in your security architecture. This high availability configuration provides failsafe inspection of network traffic to keep your defenses strong.

High availability with near-instant failover

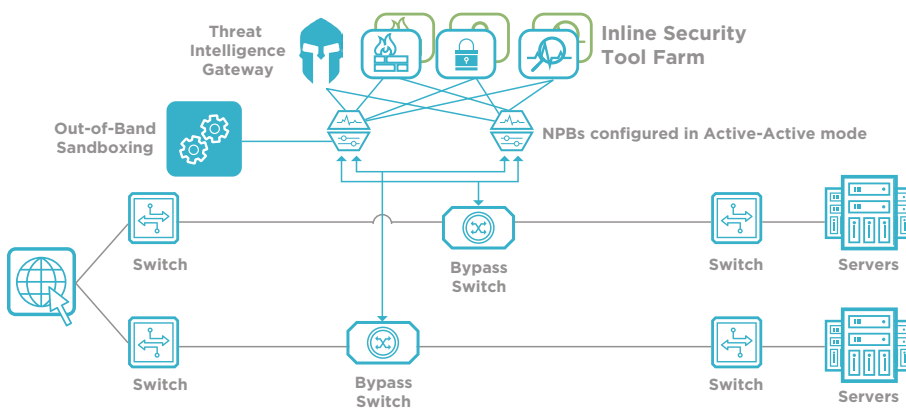


Figure 5. High availability security architecture for near-instant failover

Even when a standby device is available, the time it takes to failover can be unacceptable in environments such as financial services where delays can impact financial results. For organizations like these, some NPBs can be configured in active-active mode, where both devices are actively processing production traffic at less than full capacity and are configured with complete

synchronicity and session awareness. The key is that the capacity of one NPB must be enough to handle 100% of normal traffic in the event the other NPB goes offline. Active-active processing offers a significant improvement in throughput during normal operations and sub-second failover, so no packets are lost during the transition.

CONSIDERATIONS FOR CREATING HIGH AVAILABILITY

Share normal workload

The cost of redundant architecture can be a substantial portion of the overall security budget. Many organizations find it easier to justify the cost of a second NPB if it can be used immediately in normal operations, sharing the workload with the primary device through an active-active configuration. Periodically switching packet processing to a single NPB is a good way to prove that the existing configuration is still able to handle the entire volume of traffic.

Eliminate risk of incomplete failover

With a secondary device in standby mode, there is some risk it will fail to boot up and will be unable to take over processing if the primary device fails. With both NPBs actively processing data, there is no risk of an incomplete failover.

Enable maintenance without disruption

With an active-active configuration, any maintenance that needs to be done on an NPB or any of your security solutions can be performed without the need for a maintenance window. Traffic can be temporarily routed through a single device until both devices are back online.

MORE INFORMATION

Find out more about how to achieve complete security resilience with Ixia iBypass switches and Vision ONE packet brokers:

- Solution brief: [Creating Resilience Within a Security Architecture](#)
- Solution brief: [High Availability Makes Monitoring More Reliable](#)
- Case study: [High Availability Security Upgrade](#)

Inline Security Monitoring

When it comes to network security monitoring, there are two deployment scenarios: inline and out-of-band. These terms refer to the placement of the security appliance in the flow of network traffic. A device in the path of network traffic as it traverses to its destination is referred to as an inline (or occasionally in-band) device. If the device is not in the path of live traffic, but is processing copies of live traffic, or mirrored traffic, then it is referred to as an out-of-band device.

PURPOSE OF INLINE MONITORING

The type of security monitoring determines whether an appliance is deployed inline or out-of-band. Monitoring which is done in real-time on live traffic, to prevent malicious actors from entering or leaving the trusted network with sensitive data, is called inline monitoring. Unfortunately, the process of inspecting and fully analyzing network traffic can take time and introduce a bit of delay in the response to users. Therefore, organizations must balance the need for responsiveness and network availability with the desire to inspect traffic. The result is usually a combination of inline security monitoring and out-of-band analysis conducted on mirrored traffic. For instance, firewalls are located inline at the edge of the organization's network interface to the outside world to block known bad traffic. On the other hand, security forensics solutions that employ complex algorithms to find suspicious patterns in network traffic are deployed out-of-band.

Security appliances deployed inline must be protected from affecting the network by deploying a bypass switch in front of the inline device.

Here is a simple example of inline security:

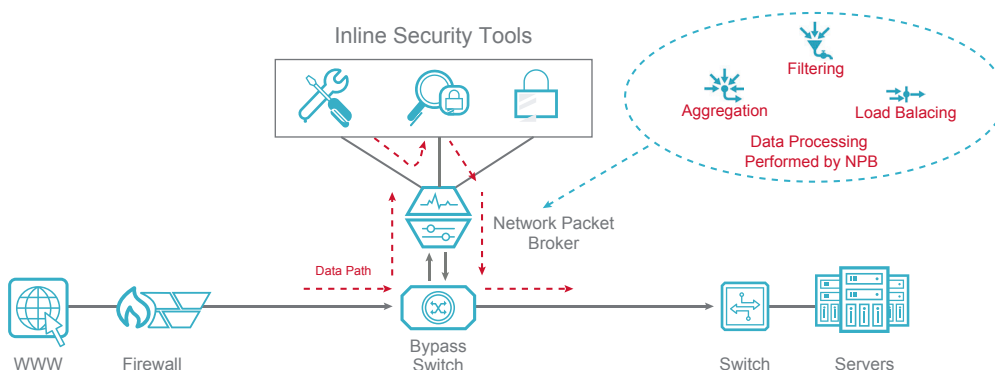


Figure 6. Inline security monitoring system

USE CASES AND BENEFITS

Threat Intelligence

It is becoming more common for organizations to deploy some type of solution that uses current threat intelligence to separate out known bad traffic before it even arrives at the firewall. These solutions are deployed inline and help your security appliances work more effectively by pre-filtering traffic that is known to be suspicious at line rate speeds.

Firewall

A firewall establishes a barrier between a trusted internal network and an untrusted outside network, such as the Internet. The firewall compares each packet received to a set of established criteria to determine if a packet will be forwarded. Packets that do not satisfy the filter criteria are either dropped or sent to another area outside the network for further analysis.

Intrusion Prevention Systems

While the firewall works to positively identify traffic that is allowed to move on towards the internal network, the IPS looks for dangerous incoming packets or traffic that violate specific rules or network policies. Once suspicious traffic is identified, the IPS takes action by automatically blocking the traffic, logging the attack, and adding the source IP address to the block list for a period of time. IPS systems can also identify port scans that hackers use to find a vulnerability in a particular network.

Anti-Virus Gateway

Antivirus software has evolved to become a multi-tiered solution running everywhere from the desktop to the firewall, to mobile devices. Most commercial-class firewalls scan content for viruses, spyware, and other malware.

CONSIDERATIONS FOR DEPLOYING INLINE SECURITY SOLUTIONS

Here are some things to keep in mind when planning your security architecture:

Monitoring Goals

Inline security solutions placed in the direct path of live traffic moving towards its destination are focused on the inspection of individual packets to determine whether the packet should be passed into or blocked from entering the trusted network. Firewalls are the most common inline solutions and are deployed by nearly every company—large and small—to block suspicious traffic based on content, source, destination, or type. Other inline solutions include intrusion prevention (IPS), data loss prevention (DLP), antimalware, web application firewall





(WAF), and malware sandboxes. With increasing complexity in security attacks and the rise of multi-layer attacks, the best security architecture is generally a combinations of pre-connection traffic inspection and post-connection threat analysis performed by out-of-band tools.

Performance

Inline security solutions, and the visibility architecture that supports them, are considered critical to functioning of the enterprise network. Performance is paramount. A visibility platform can relieve pressure on security solutions by filtering and processing raw data and delivering only the data each solution needs. Your taps, bypass switches, and packet brokers should process data at line rate under full load, without losing packets or becoming congested. It is also prudent to test each new security component under realistic traffic conditions prior to deployment.

Scalability

Scalability is also important since over-subscribed tools that do not keep up with traffic growth can affect performance and increase management costs. Network packet brokers (NPB)s with automatic load balancing can relieve overloaded security tools by distributing traffic over multiple devices, to increase efficiency and reduce congestion. New security tools can be easily added to an established group by letting the NPB automatically distribute traffic among all available inline solutions.

Extend Life of Security Tools

A visibility platform also separates your security monitoring tools from the production network, to allow lower-speed monitoring tools to process data from higher-speed network links. This lets you make independent decisions about when to upgrade your network devices and security solutions and extend the useful life of lower speed security appliances.

MORE INFORMATION

Find out more about preventing security attacks on your network:

- Blog: [Take the Fear Out of Real-Time Security Monitoring](#)
- Solution brief: [Strengthen Security and Reduce Risk of Downtime](#)
- White paper: [Evaluating Inline Security Fabric: Key Considerations](#)

Intrusion Prevention (IPS) and Detection (IDS) Systems

Like many things in the rapidly changing world of IT security, the functions of these systems are not as distinct as they once were. Both types of security monitoring solutions look for intrusions and violations of network policy, in order to protect an organization from being compromised. Traditionally, however, they were designed to be deployed in different parts of the network and focus on different goals.

FUNCTION OF AN IPS

Intrusion prevention was designed to be deployed inline on the network, close to the perimeter, and complement the work of the network firewall. While the firewall works to positively identify traffic that is allowed to move on towards the internal network, the IPS looks for dangerous incoming packets or traffic that violate specific rules or network policies. Once suspicious traffic is identified, the IPS takes action by automatically blocking the traffic, logging the attack, and adding the source IP address to the block list for a period of time. IPS systems can also identify port scans that hackers use to find a vulnerability in a particular network.

As an inline device, the IPS must perform its inspection work quickly to avoid degrading network performance and to stop potential attacks in real-time. The IPS must also detect and respond accurately, to eliminate false positives or alerts that must be followed up by security staff.

FUNCTION OF AN IDS

Intrusion detection is used offline, or out-of-band, to identify and log violations and send an alert to an administrator or report the violation to a central repository called a 'security information and event management (SIEM) system.' A SIEM system can centrally combine alerts from multiple tools or sources to better distinguish malicious activity from false alarms. The traffic that is sent to an IDS is a copy of live traffic, generated by a SPAN port or network tap, and is not routed back into the trusted network. This is sometimes referred to as passive monitoring, since no automatic action is taken. Because it does not operate on live traffic and have the constraint of having to perform at line speed, an IDS can be used to perform more complex analyses and investigations.

IDENTIFICATION TECHNIQUES

The providers of IPS and IDS systems continually develop new ways to identify threats and circumvent security breaches. Initially, these systems relied on signature-based identification, in which past attacks were analyzed to come up with identifying characteristics (or signatures) that the appliances then search for. The limitation, of course, is that new attacks need to be successfully identified and characterized before they can be added to the search criteria. Statistical anomaly-based techniques were then added so the systems could produce alerts based on traffic that was deemed out of the ordinary. This helps to flag what could be new attacks, but also requires a fair amount of system tuning to limit the number of false positives. Many IPS and IDS systems combine signature and anomaly-based detection.

More recently, rule-based techniques are being used to go beyond simple packet inspection and make more sophisticated predictions based on multiple events taking place on the network. The idea is that if Event A and Event B both take place, neither of them necessarily suspicious on their own, but are then followed by Event C, then it can be presumed an attack is underway. This capability is sometimes referred to as an “inference engine” and can help preempt attacks.

CONSIDERATIONS FOR DEPLOYING INTRUSION PREVENTION

Fail-Safe Operation

Since IPS devices are deployed inline on the network, they have the potential to cause network disruption should they experience an outage. Every device deployed inline, behind the firewall, needs to be made fail-safe to preserve network uptime. A [bypass switch](#) performs this function by continually checking on the status of your IPS and confirming its ability to receive and process traffic (inline mode). If your IPS goes offline for any reason—such as a power or port failure, a software configuration error, or congestion caused by a burst in volume—the bypass switch automatically routes the traffic around the appliance (bypass mode), to keep applications and services responding.

Zero Downtime Maintenance

In addition to failing open, an external bypass also allows administrators to proactively bypass the inline device like an IPS, to perform maintenance functions without waiting for a network maintenance window. Some administrators refer to this as bypass mode, where traffic is proactively routed around your IPS during troubleshooting or repair. This is something an internal bypass function is not able to do.

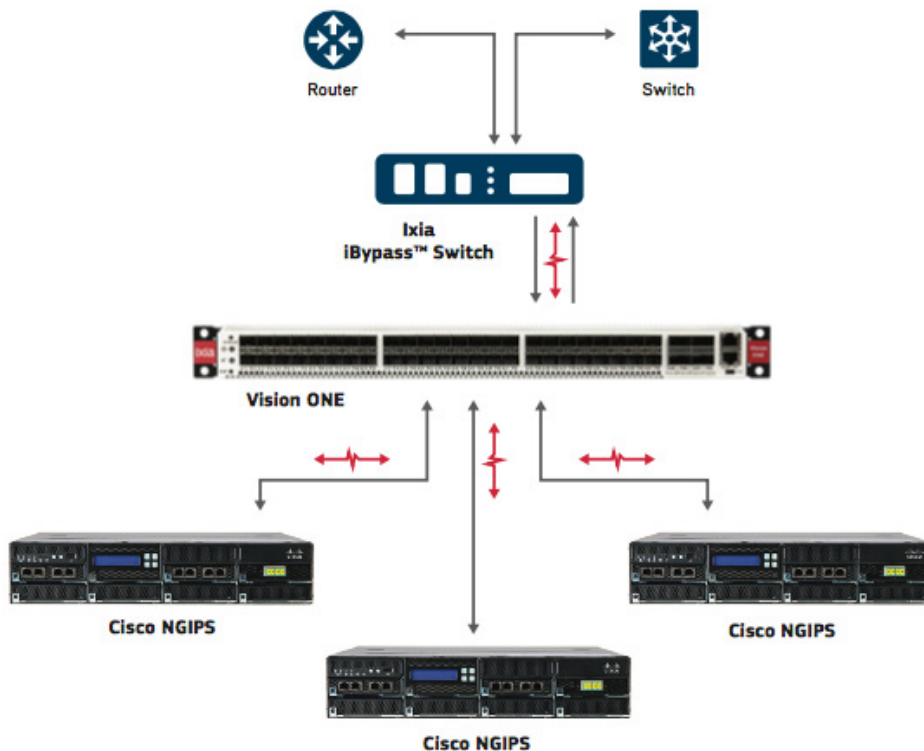


Figure 7. Failsafe inline deployment of Cisco intrusion prevention systems

Accelerated Deployment

Deploying an IPS is not a trivial task and the tuning and configuration of a new IPS generally takes at least several hours to complete. To avoid bringing the network down for that long, an external bypass switch can be installed in bypass mode and connected to the new IPS fairly quickly. Since traffic is being routed around the device, this give security administrators the time they need to configure and test the IPS, without impacting network traffic.

Out-of-Band or Tap Mode

If your organization is new to security monitoring with an IPS, you may want to build confidence using the device before activating it in your live network. Some external bypass switches, such as Ixia's iBypass, not only allow your IPS to be in inline (inspection) mode and bypass (maintenance) mode, but also in "tap mode," which means live traffic arriving at the bypass is routed through to the network and a copy of the traffic is sent to your monitoring device. With a copy of the live traffic, you can come up-to-speed using the IPS before switching it to inline mode. An IPS operating in tap mode is essentially functioning as an IDS. In a recent discussion with one of Ixia's sales engineers, I learned that a lot of customers leave their IPS in tap mode, apparently appreciating the functionality of the IPS and the flexibility of knowing they can switch to inline monitoring at some point in the future.

Visibility

Attackers look for places to embed code that might be overlooked by your security monitoring devices. Make it harder for them by inspecting as much traffic as you can. Install a security fabric to aggregate traffic from across your network and make it available for inspection and detection.

MORE INFORMATION

Find out more about keeping your security solutions safe and effective:

- Solution brief: [Get More Value from Security and Monitoring Tools](#)
- White paper: [Evaluating Inline Security: Key Considerations](#)

Malware Sandbox

Advanced malware and advanced persistent threats (APT) now threaten businesses of all sizes and have led to more sophisticated detection techniques. Many organizations turn to a malware sandbox, a type of behavioral analysis, to observe the impact of executing a particular command or program. Once used exclusively in manual research, sandboxes are now also routinely used for automated detection. Using a malware sandbox is more effective than other security methodologies because it returns a verdict based on what the suspected malware actually does, rather than how it appears.

FUNCTION OF MALWARE SANDBOX

Sandboxing works by running code (sometimes referred to as “detonating code”) inside a tightly controlled environment, where analysts can observe what actually happens, rather than trying to anticipate malicious behavior based on attributes or signatures. Sandboxes help analysts identify attempts to command and control internal servers, the presence of persistence techniques, or attempts to compromise the operating system. Since it is not necessary to have previously identified a specific threat, sandboxing offers the ability to identify zero-day threats.

Because malware analysis requires more time than other threat detection solutions, it is important to pre-filter the traffic to avoid wasting time analyzing “known good” or “known bad” solutions. Often, the sandbox is one component of an overall advanced threat protection (ATP) solution.

While many sandboxes are passive and can only report incoming threats, some solutions may be integrated with a next generation firewall (NGFW) or unified threat management system (UTM) deployed inline to proactively pre-filter malicious traffic out of the network. Some vendors supply solutions that provide both automated analysis and an active threat response.

USE CASES AND BENEFITS

Advanced threat protection

A malware sandbox is one component of a broader advanced threat protection strategy. Cyber threats have become too sophisticated for point products or traditional tools and require a multi-faceted approach. Nation-states with tremendous resources and strong motivations are engaged in developing advanced malware that can evade detection. Advanced threat protection has evolved to address this environment.⁴

⁴ Frost & Sullivan: Advanced Malware Sandbox Market Analysis, September 2016.

Reduce dwell time

Dwell time begins when an attacker breaches the network and continues until the attacker leaves on his own or is detected and kicked out. According to the 2016 Verizon Data Breach and Investigation Report, the average dwell time is 200 days, which leaves the attacker plenty of time to steal data or credentials, wipe out evidence of his presence, and set up a back door to make future attacks more convenient. Reducing dwell time, therefore, does more than limit damage; it can also help prevent future attacks. Sandboxes can accelerate the time it takes to detect and isolate malicious malware.

Increase team productivity

The purchase of a malware sandbox is often justified by its ability to reduce the number of false negatives (malicious code that evades security solutions and potentially leads to an attack) and the number of false positives (security alerts that must be researched by staff). Both situations can end up costing the security team time and effort. Alerts confirmed as malicious in a sandbox can then be prioritized for the security team to follow-up, also increasing productivity.

High-security environments

Used in combination with threat intelligence solutions, malware sandboxes are often deployed in sensitive environments, including government security, defense and intelligence agencies.

CONSIDERATIONS WHEN DEPLOYING A MALWARE SANDBOX

Free or Fee-based

There are many free, cloud-based malware sandboxes, but it is easy for malware perpetrators to use the same free sandboxes to design attacks to specifically evade these tools. You will get much greater value and protection from a sandbox with a highly-sophisticated analysis engine and the ability to analyze Windows, Apple, and Android communications.

The open source community has also responded with alternatives to commercial offerings. These alternatives, believed by many to have a lower cost, require a considerable investment in time and expertise. These solutions are not in the same category as the free tools available online.



Dual-Detection and Superior Analysis

There are two different ways to design a sandbox. One way is to emulate the different target environments where the code will be executed. The second way is to replicate the actual production environments using virtualization. The best sandbox solutions combine both approaches to have the best chance of observing the malicious behavior. In addition to executing the code, advanced malware solutions may also include statistical analysis using algorithms and machine learning to determine patterns that are similar to known threats.

Ability to Avoid Detection

In the profitable world of cybercrime, malware is continually being adapted to sniff out the presence of a sandbox and adapt the code to avoid detection. Best-in-class sandbox solutions use a combination of methods to unmask disguised malware that alter their behavior when they detect the existence of sandboxing techniques. These solutions are sometimes referred to as “advanced malware sandboxes.”

Reporting and Real-Time Alerts

Real-time incident reporting with detailed analysis provides immediate notification to security analysts. Many solutions supply independent risk scores and confidence statistics. Details are recorded to help staff understand the nature of emerging threats and identify trends.

Integration with Visibility Platform

As with first generation security solutions, a malware sandbox must have access to all of the relevant network traffic to adequately protect an organization from advanced malware. Most sandboxes will not have the ability to decrypt SSL traffic and a visibility solution with that feature can perform that function.

Integration with Other Security Solutions

As networks evolve and cloud technologies become more pervasive, organizations may want to deploy sandboxes with the ability to provide actionable intelligence and active threat response. Solutions designed to automatically share information with threat intelligence solutions will provide the fastest response to newly-exposed malware.

MORE INFORMATION

Find out more about preventing security attacks on your network:

- Report: [Security Report 2017](#)
- Webinar: [Bringing Evasions Out Into the Open](#)

Network Packet Brokers

Having the right monitoring and security tools has never been more vital. Organizations often end up using a blend of tools from different vendors. But trying to manage a mix of tools gets complicated very quickly. And there's no room for error—since monitoring and security tools are only as good as the data they see. Network packet brokers are a practical way of addressing this problem, and strengthening network security.

WHAT IS A NETWORK PACKET BROKER?

A network packet broker (NPB) is a device that provides a collection of monitoring tools with access to traffic from across the network. The diagram below shows how an NPB receives data from a number of network links. It then acts as a “broker” distributing the relevant data out to the monitoring tools. NPBs are active brokers because they can filter the data that is supplied to each tool.

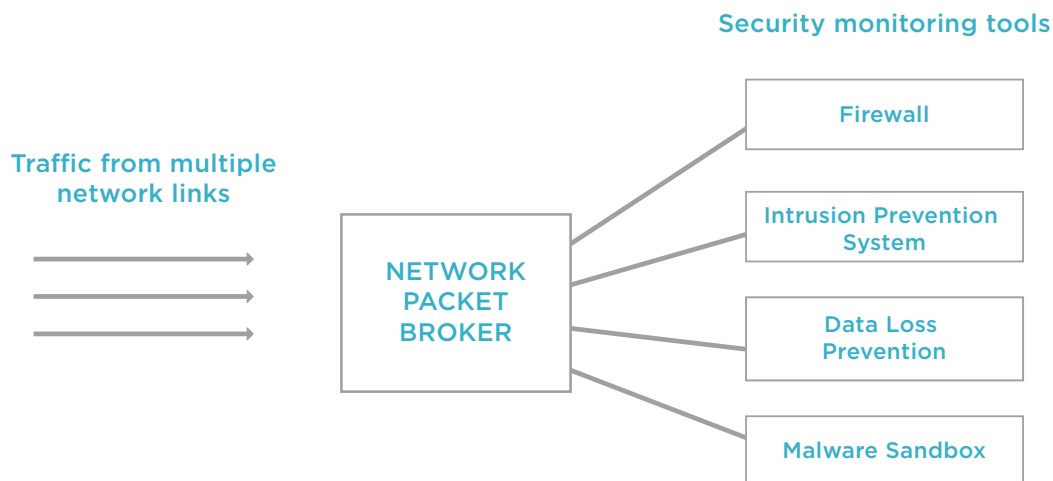


Figure 8. Network packet broker

NPBs can:

- Distribute data from one network link, to one tool
- Distribute data from one network link, to multiple tools
- Distribute data from multiple network links, to one tool
- Distribute data from multiple network links, to multiple tools

Ultimately, NPBs make monitoring and security tools more effective, by giving them access to a range of data from across the entire network. Blind spots are reduced, giving tools the visibility they need to identify and tackle performance and security threats.

USE CASE AND BENEFITS

Maximize return on security investments

Many of the expensive security appliances and monitoring tools deployed on a particular network segment are not fully utilized, which limits the value they provide. An NPB offers a way for the capacity of these solutions to be shared across the entire enterprise and increase return on investment. NPBs allow you to aggregate traffic from multiple network links and deliver it to a single security appliance.

Increase security tool efficiency

Not all network traffic needs to be inspected by every one of your security tools. As the volume of traffic entering your organization increases, your security appliances may become overloaded. One way to improve efficiency is to use a packet broker to remove out duplicate packets collected from multiple links and filter out low-risk traffic such as voice or video that does not need to be inspected. Filtering reduces the load on your security appliances and help you delay new purchases, to stretch your budget further.

Enable high availability and security resilience

A security breach can have a devastating impact on an organization. Most organizations require high availability for their security infrastructure, as they do for their core network. By pairing redundant external bypass switches with redundant NPBs, you can achieve very high uptime for your security monitoring. Configuring the primary and secondary NPBs with complete synchronicity in active-active mode gives you near-instant failover.

Improve threat detection and prioritize alerts

Following up on security alerts can be a major drain on productivity for your IT security team. Alerts that are not investigated can lead to data loss, but keeping up is daunting. One way to prioritize alerts is to automatically pass suspect packets through a series of security tools, referred to as a serial tool chain. NPBs let you specify the route of a packet from one tool to the next, depending on the result, to reduce the number of unresolved alerts and help you focus on the alerts that are most serious.



CONSIDERATIONS FOR EVALUATING PACKET BROKERS

Network packet brokers aren't all made equal. NPBs are a fairly new form of technology and vendors vary in the way they design these products.

High Performance Data Processing

Your security and visibility architecture must operate with high performance, particularly as core network speeds move from 10Gbps to 40, 100, and beyond. Equally as important is the ability to handle fluctuations in traffic load, with zero packet loss. Unfortunately, some organizations are forced to make trade-offs in the type of filters and packet processing they do, because their processing engines are not up to the task. Look for NPBs with optimum flexibility and the ability to process packets with all the features you need at realistic loads.

Context-Aware Filtering

Managing the network means knowing what's on the network. Large networks can have hundreds of applications and many thousands of endpoints, especially with the growth in BYOD. Intelligent NPBs can identify the applications or devices active on the network and provide that information to your security and monitoring tools. This makes your monitoring and security tools much more efficient, and it makes life much easier for administrators.

Onboard SSL Decryption

Secure Socket Layer (SSL) encryption is the standard technology used to send private or sensitive information. While SSL hides sensitive data, it can also end up encrypting and hiding malicious cyber threats too. To maximize security, [organizations should decrypt and examine SSL traffic](#). The difficulty is that decryption takes valuable processing power. If decryption is performed by your security tools, they have less cycles left to inspect packets and keep your network safe from threats.

A more efficient alternative is to have an intelligent NPB perform SSL decryption and pass the decrypted data on to all your monitoring tools. This gives them the ability to see all encrypted traffic without having to spend the processing cycles to do the decrypting.

Data Masking

One drawback to SSL decryption is it makes all data visible to anyone that has access to your monitoring tools. Some of this unencrypted data may be quite sensitive and protected by regulatory requirements. Data like personally identifiable information (PII), or credit card information must be protected and not exposed to unauthorized individuals. Thus, advanced NPBs can mask sensitive data that should not be exposed to monitoring and security tools or their administrators. Data masking can be a critical NPB feature that makes monitoring activities safe.

Simple Management Interface

As you configure connections and define filters, management and control of your visibility platform can become complex, especially when you have many traffic sources and many tools. Command Line Interfaces (CLIs) quickly become cumbersome and prone to errors and mistakes that you may not uncover until something goes wrong. Therefore, NPBs with a centralized, drag-and-drop interface will simplify configuration and management of visibility to your security solutions. You have complete granular control and the ability to update filters as your needs change.

MORE INFORMATION

Find out more about the advantages of Ixia Vision ONE packet brokers:

- E-book: [Network Visibility For Dummies](#)
- White paper: [The Importance of Lossless Network Visibility](#)
- Solution brief: [NPB Automation Improves Security Monitoring](#)

Next Gen Firewall

The emergence of web applications and the expanding sophistication of security threats have driven an evolution in firewall technology. Application-level monitoring must be performed in addition to analysis of port and destination and next generation firewalls (NGFW) have evolved to address this need. In their 2017 Magic Quadrant for Enterprise Network Firewalls, Gartner reports that “next generation” capabilities have been achieved by all products in the firewall market and vendors differentiate only on feature strengths.⁵

FUNCTION OF A NEXT GEN FIREWALL

NGFWs must be capable of performing deep packet inspection on all packets, on all ports, and over all protocols in order to secure applications effectively. In addition, with the expanded use of SSL/TLS in traffic traversing the internet, inspection of encrypted content is also required. NGFW features commonly include: (1) the ability to enforce policy via application and end-user controls and (2) the ability to detect new threats using some combination of intrusion prevention, sandboxing, and threat intelligence feeds. Since NGFWs are deployed at critical choke points in the network, their stability and reliability is imperative.

Next generation firewall features (as opposed to traditional firewalls) include⁶:

- Application awareness/control
- Identity awareness/control
- Integrated intrusion prevention system (IPS)
- Ability to use external intelligence sources to enhance blocking (threat intelligence)
- Anti-malware
- SSL decryption and inspection

USE CASES AND BENEFITS

Expand protection to internet edge

Sophisticated malware requires that firewall protection be extended beyond the network perimeter to include every networked device, as well as collaboration tools, streaming media, and cloud applications.

⁵ Gartner “Magic Quadrant for Enterprise Network Firewalls,” 12 July 2017.

⁶ NSS Labs: “Test Methodology for Next Generation Firewalls,” 2016, [accessed online](#).

Eliminate separate intrusion prevention systems

With the evolution and advancement of firewall technology, many enterprises are gradually replacing stand-alone network intrusion prevention systems (IPS) at the enterprise edge and the Although the functionality of today's enterprise firewalls generally negate the need for a separate NGFW, Network Security.

Augment existing firewall with additional layer of security

Many organizations choose to leave their existing first-generation firewalls in place at the network edge and deploy a NGFW behind that device for its advanced features. This configuration allows them to lengthen the service life of existing security infrastructure and reserve the capacity of their new advanced devices for more sophisticated filtering and monitoring tasks.

Increase security resilience

Many organizations choose to leave their existing first-generation firewalls in place at the network edge and deploy a NGFW behind that device for its advanced features. This configuration allows them to lengthen the service life of existing security infrastructure and reserve the capacity of their new advanced devices for more sophisticated filtering and monitoring tasks.

CONSIDERATIONS WHEN DEPLOYING NEXT GEN FIREWALLS

Latency

Given the intense processing and analysis of packets, traffic latency can be a major concern. Many devices deliver sustained speeds of 10 to 40 Gigabits or more, but these should be tested with realistic traffic loads prior to live deployment to ensure service levels can be maintained. Make sure to you're your device with all of the key features turned on. Many NGFW experience significant latency with SSL decryption and encryption enabled. A better result may be obtained by offloading decryption to a separate device or to a visibility platform.

Stability and Reliability

Long-term stability is critical for an inline device where failure can produce network outages. NSS Labs recommends verifying the stability of the NGFW while under normal loads and while passing malicious traffic. Products that are not able to sustain legitimate traffic, or that crash, while under hostile attack will not be adequate to their intended purpose.



Zero Downtime Deployment and Maintenance.

With an external bypass switch already deployed, it is possible to deploy a NGFW without downtime. Setting an external bypass switch “on” allows traffic to continue flowing through your live network while you install and configure your NGFW behind the bypass. Once deployment of the NGFW is complete, you can turn the bypass “off” and live traffic will begin flowing through the NGFW. The same process works for upgrades to the device. If an external bypass is not already deployed, you can deploy it along with your NGFW. It takes only minutes, allowing you to minimize downtime during deployment.

Load Balancing for High Availability

Load balancing ensures high availability for all the services behind the firewall. Having extra capacity and being able to shift workloads between devices helps ensure your organization’s web-based services are always available. A network packet broker (NPB) deployed in front of your inline security devices can provide automatic load balancing that you can control from a centralized user interface. NPBs also enable easy scaling, since they can automatically discover newly attached devices and use pre-determined rules to begin load balancing incoming traffic.

MORE INFORMATION

Find out more about how to support your next generation firewall deployment:

- White paper: [Plan Your Firewall Migration to Increase Security Resilience](#)
- White Paper: [Taking the Risk and Cost Out of Security Upgrades](#)
- Solution brief: [Expandable Next-Gen Firewall Security](#)

Service Chaining

Service chaining is a powerful tool for automating the movement of data packets and it's getting a lot of attention these days as a way to improve the quality and speed of application delivery.

PURPOSE OF SERVICE CHAINING

A 'service chain' is a set of services which are performed in a specific order and 'service chaining' refers to steering the traffic through such a chain. It's like a recipe where actions are performed in a pre-ordained order. Services can be performed in parallel or in serial, depending on the situation. The chain can be implemented by cabling individual devices together or, increasingly, by using software provisioning to control the flow of data through the selected services. Monitoring tools that are linked together in this way are sometimes referred to as a daisy-chain.

The use of service chains is linked to the automation of functions that have been either embedded in single-purpose hardware devices, dictated by physical topologies, or performed manually—which are increasingly perceived as too costly and inflexible in our fast-moving digital economy.

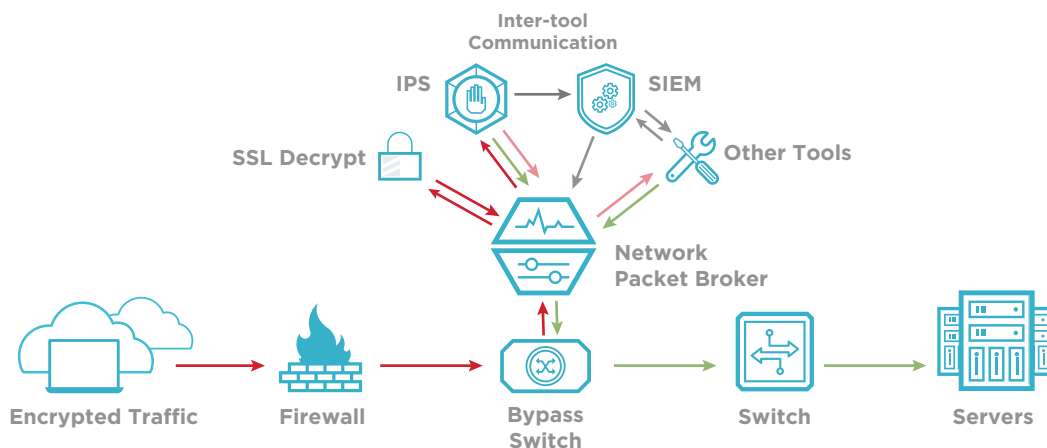


Figure 9. Example of visibility architecture using serial tool chaining

SERVICE CHAIN USE CASES AND BENEFITS

Service chaining is one of several approaches that make it possible to centrally manage and direct the operation of IT resources, to increase efficiency and time-to-market, as well as decrease costs.



Real-time network monitoring

With real-time monitoring, you need to keep traffic moving quickly and your security tools working efficiently. Chaining tools together allows you to pass only the suspicious traffic to additional tools for deeper inspection or to a honeypot to be quarantined. Packets without anomalies are moved along quickly, to maintain maximum response time. A common example is the use of a Security Information and Event Management (SIEM) solution to filter out suspicious traffic for further analysis by other tools in the daisy-chain. Traffic without exception is quickly sent back through the network to support the fastest possible response time.

Out-of-band monitoring

Out-of-band monitoring tools can be chained for similar reasons. An example would be to take the result of deep packet inspection provided by an Ixia network packet broker and send the application-specific information on to the best tool for analyzing a given packet type. Meta data can also be added to the packets to let tools farther in the chain know more about the origin or destination of the traffic.

Value added traffic management

Service chaining is also common when administrators must enable multiple resources or processes to be used. Examples are to enforce policies, perform QoS monitoring, to gather real-time analytics for traffic flow adjustments, are enforced to ensure quality of service.

CONSIDERATIONS FOR DEPLOYING SERVICE CHAINING

Enable Network Function Virtualization (NFV)

Once upon a time, specialized network appliances ruled the data center and in many places they still do. When you consider their purpose, however, you can identify multiple functions taking place inside each appliance. For instance, a firewall might perform network address translation, deep packet inspection, and access control. The hardware appliance was designed to perform these functions at wire speed. But in recent years, many of the functions once performed by expensive hardware appliances are being redesigned as software functions that can be run on any generic and low-cost CPU. This process is called network function virtualization and the goal is to achieve the same results as the appliance, but at greater efficiency and less cost.

Reduce Latency

In order to get acceptable performance in a virtualized environment however, services that run in software on a generic CPU must be chained together, to accelerate total processing speed or latency. Any time services are grouped together in a way that forces processing to proceed from step one to step two, latency can be reduced and speed accelerated.

Reduce Redundant Inspections

Without the ability to chain together certain functions, a particular packet may need to pass through a particular service more than once to meet the qualifications for other types of inspection tools. For instance, in the case of security monitoring, SSL traffic can pass through a powerful decryption tool and the exposed content can be sent through a series of additional inspection tools. This avoids the need to send the traffic through decryption for each tool, which would increase latency and multiply the cycles being consumed on the decryption tool. A more efficient and more cost-effective result is achieved by sending decrypted traffic through multiple tools before passing it through to the trusted network.

Apply Consistent Policies

Pre-set service chains help ensure that actions are taken in a specific sequence and nothing is overlooked. This reduces errors and increases the chance that abnormalities will be identified in time to prevent damage to an organization's data or other resources.

Increase Flexibility

The ability to define service chains dynamically, based on the user, device, location, service level, or other characteristic is a powerful capability in the fast-moving digital economy. Well-defined rules and policies can help decrease the time to deliver a service and increase the quality of the user experience.

MORE INFORMATION

Find out more about the use of service chaining in network security:

- Solution brief: [Improve Security Threat Analysis by Using Serial Tool Chaining](#)

SSL Decryption

Coded, disguised forms of communication (also known as ciphers) are nothing new. For centuries, people have sent messages designed to be unreadable, if intercepted or received by hostile forces. The enigma machines used by Germany during WW2 are a famous example of this.

PURPOSE OF SSL

Today, SSL (secure sockets layer) encryption and decryption are the means by which sensitive data is safely transmitted – and protected from prying eyes – over the internet and across networks.

On a basic level, SSL encryption occurs when sensitive data is transformed into an unintelligible, unreadable “ciphertext”. SSL decryption occurs when this “ciphertext” is returned to its original format. To do this, a key, which gives instructions on how to decode the encrypted message, is required.

USE CASES AND BENEFITS

Exposing encrypted security attacks

We live in an age where the stakes are high for both individuals and organizations that fall victim to data theft. So it's for good reason that SSL encryption has soared (and continues to soar) in popularity. According to Mozilla, half of all internet traffic is now encrypted. SSL encryption is a powerful weapon in the battle for data security, but its greatest strength, is also its greatest weakness.

Encryption can also hide malware and other cyber threats, so they're able to sneak into and around company networks undetected.

Since most network tools can't inspect SSL encrypted data, it's necessary that data is first decrypted, and then inspected.

Firewall strain

Many organizations use their Next Generation Firewalls (NGFWs), (which typically come with decryption capabilities) as the main point of SSL decryption on the network. But, as data encryption continues to grow in popularity, firewalls will experience more and more strain. SSL decryption takes up valuable processing power, making the firewall less effective. This makes it easy for the firewall to become a bottleneck on the network, holding up other processes and stalling productivity.

Internal threats

Following on from the above, firewalls often sit at the edge of the network, only decoding encrypted data that comes from outside. This means that encrypted communications that occur internally (between servers / clients) remain uninspected. This is a huge risk, since internal communications may comprise some 80% of encrypted network data. If malware does somehow make it into the network, SSL encryption within the network will help camouflage it.

Inefficiency and overloading

Some network monitoring tools (aside from NGFWs) come with SSL decryption capabilities too. This isn't an ideal solution however. As with firewalls, enabling SSL decryption on these tools can debilitate performance. Furthermore, requiring each tool to decrypt its own data is inefficient. It means multiple, siloed tools, performing the same decryption process, on the same set of data. This is a waste of resources. Why have several appliances repeating the same task when one tool could decrypt the data, and then push it out to them all?

Data compliance

HIPAA, PCI, and organizational best-practice policies mean personally identifiable information (PII) must be handled with care. When SSL decryption occurs, sensitive, personal data is exposed. Without the right safeguards in place, anyone with access to network monitoring tools becomes privy to this sensitive information. Data masking should be used to protect this data.

CONSIDERATIONS FOR HANDLING ENCRYPTED DATA

Companies can overcome the challenges associated with SSL decryption by using a network packet broker (NPB) with decryption capabilities. Network packet brokers act as a central recipient of and “dealer” of data in the network. By decrypting the data at this point, before sending it out to network appliances (e.g. Intrusion Detection Systems (IDS), Intrusion Prevention Systems (IPS), Firewalls), the NPB removes the processing burden from these tools, boosting overall efficiency and productivity. Unlike firewalls, NPBs are also able to decode encrypted data that arises from within the network.

MORE INFORMATION

Find out more about the use of service chaining in network security:

- White paper: [Inspecting SSL Traffic](#)
- Solution brief: [Security in an Encrypted World](#)
- Solution brief: [Role of Active SSL Decryption in Network Monitoring](#)

Threat Intelligence

Threat intelligence can be defined as evidence-based knowledge about existing or emerging security threats, threat actors, exploits, malware, vulnerabilities, and compromise indicators that can be used to take action in defense of an enterprise. Solutions include hardware devices known as intelligence gateways, real-time feeds providing the latest information on active and emerging threats, and security services that provide recommendations on how to increase the level of security.

FUNCTION OF THREAT INTELLIGENCE

Threat intelligence protects an organization from cyberattacks in a way that differs from other security appliances. The value of threat intelligence comes from its ability to reduce your network's attack surface and simultaneously help your security appliances work more effectively. Threat intelligence is based on investigative and verification work performed by experienced security analysts. Their findings are maintained in a database that can be used to pre-filter traffic that is either attempting to enter or leave your network. A threat intelligence gateway is a hardware device that is designed for the sole purpose of using the intelligence to block suspicious traffic at line rate speeds.

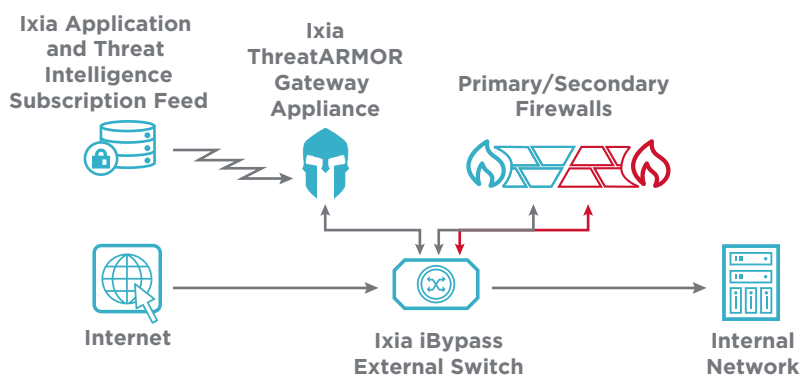


Figure 10. Failsafe implementation of threat intelligence solution

USE CASES AND BENEFITS

Reduce the attack surface

When you're dealing with network security, one of the primary measurements of risk is the size of your "attack surface." Most hosts on the Internet are legitimate or safe for your organization to communicate with. They are just people going about their business. However, because IP addresses can be difficult or expensive to acquire, they are often reused to launch malware, botnets, phishing, or other cyberattacks. These sites don't belong on your network. There are also unallocated IP addresses that don't belong to anyone and also shouldn't have access to your network. And there may be regions or countries in the world where you don't do business and don't want hosts from these areas communicating inside your network. Being able to filter out traffic from these sites is known as reducing the attack surface and minimizes the chance of a successful attack against your organization can be attacked.

Reduce security alerts

A threat intelligence gateway is typically positioned adjacent to an edge router, in front of the firewall and other security appliances. If you can block traffic that is already known to be malicious and prevent it from flowing to your security appliances, you can reduce the number of security alerts and the work your staff must to do resolve them. The effect on security alerts is immediate, particularly in organizations that are frequently targeted by cyberattacks. With fewer alerts to process, your security team can focus on unknown or emerging threats to boost overall security.

Offload your firewall to conserve its capacity

Security appliances like firewalls and intrusion prevention systems (IPS) are designed primarily for packet inspection, not for simple filtering. While your existing firewall or IPS can be configured to block traffic, this function uses up valuable processing power and requires your staff to create and maintain necessary filters. When the number of IP addresses you want to block reaches the upper limit of your firewall's range (typically 10,000 to 40,000), you will see degradation in the processing speed and accuracy of your firewall. With a threat solution deployed, your security appliances will have more processing cycles available. The extra headroom may free up enough capacity to eliminate a costly upgrade or activate an advanced firewall feature.

CONSIDERATIONS FOR DEPLOYING THREAT INTELLIGENCE

High-Performance Appliance

While a firewall or IPS can block up to 10,000--40,000 IP addresses, a threat intelligence gateway has the capacity to block an unlimited number of IP addresses with fast, dedicated processors and no degradation in processing speed or dropped packets. It is also useful if the gateway lets you create a custom blacklist to block sites that may not be flagged by the gateway, but represent a specific issue to your organization.

Automatic Real-Time Updates

Threat intelligence solutions can be designed to receive automatic, cloud-based intelligence updates every few minutes to ensure they have the latest information without the need for manual updating. This type of gateway begins working right out-of-the-box once the administrator plugs it in and connects it to the Internet.

Proven Intelligence

The key to this solution is the security research and resulting threat intelligence that feeds the gateway device. You want intelligence based on deep security research, from a team with a proven track record and rigorous policies for verifying threats. Removing sites from the database once malicious activity has been resolved is also important to avoid false negatives where legitimate communications are blocked and business is disrupted.

Documentation

Without a way of observing the actions taken by the threat intelligence gateway, it can be difficult to assess its value. Your solution should provide you with verifiable, visual documentation of all actions taken by the gateway.

Resiliency

Since disruption of any inline security appliance can harm your business, find out what safeguards are available to safeguard operation of the intelligence gateway. Dual-redundant power supplies and a built-in bypass function will ensure the gateway never becomes a point of failure in your network.

MORE INFORMATION

Find out more about the use of threat intelligence and intelligence gateways:

- White paper: [Advanced Security Intelligence with Rap Sheet Analytics](#)
- Solution brief: [Block More Threats Faster with Threat Intelligence](#)

Traffic Filtering

How do you find a needle in a haystack? Monitoring and securing modern networks requires finding “the needle” without destroying the network or the network traffic. Very sophisticated and automated analytics tools make this possible.

It is simply not possible to monitor and secure our networks without these automated tools. But these tools are expensive. How do we get the most from our monitoring and security investments? Network packet brokers (NPBs) can help. Intelligent NPBs can filter and process raw traffic and provide the results to your security monitoring solutions.

PURPOSE OF TRAFFIC FILTERING

Traffic filtering allows the user to identify the traffic they want to see and traffic they do not want to see. Filtering is designed to reduce the amount of data that must be processed by your security tools, making the individual tools much more efficient. The Ixia Vision ONE packet broker includes the filtering features shown in the graphic below.

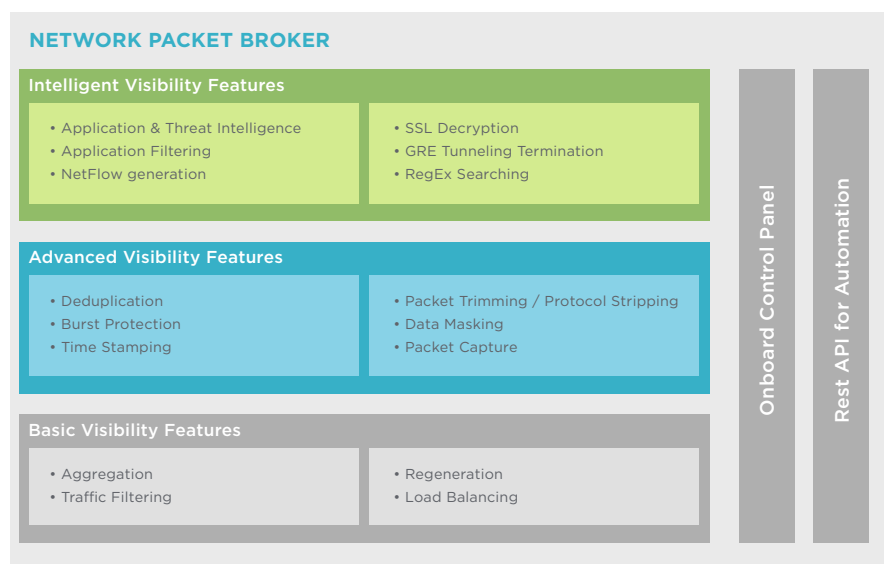


Figure 11. Intelligent network packet broker

Traffic filtering allows you to reduce the amount of tool capacity needed. For tools operating at or near capacity, that can mean less congestion and better overall performance. You may be able to delay or cancel a planned upgrade and use the saved budget dollars to purchase other tools you may not have had the funds for otherwise. In short, filtering traffic helps you make more efficient use of your security investments.

USE CASES AND BENEFITS

Reducing security tool costs

The University of Texas at Austin deployed intelligent NPBs in their network to make better use of their security budget. Using Ixia Vision ONE application filtering, they were easily able to send certain traffic, like student Netflix movies and streaming music around their intrusion prevention system (IPS). This simple step reduced the load on their security tools by 20-30% and delivered a 100% return on investment.

Identifying encrypted traffic

According to Gartner research, secure sockets layer (SSL) traffic is a significant portion of all Web traffic and is increasing. Unfortunately, encryption prevents monitoring and security tools from viewing and inspecting the traffic. With an intelligent NPB, companies can use application filtering to identify SSL traffic and send only this traffic to decryption tools.

Expediting troubleshooting

Reducing mean-time-to-resolution is a critical metric for IT organizations. Isolating certain types of traffic for forensics tools or built-in packet capture is an important troubleshooting feature on NPBs that helps significantly speed trouble isolation and reduce resolution times.

CONSIDERATIONS WHEN EVALUATING NETWORK PACKET BROKERS

Traffic filtering can be one of the most complex operations performed by a packet broker, so it is critical to know what to look for when evaluating these solutions. Below are some important NPB selection criteria.

Application Layer Filtering

Being able to easily route application flows is critical for network visibility and security. Many NPBs only route traffic based on Layer 2-4 protocols. Consider an NPB with deep packet inspection and application intelligence that can filter based on application type or packet content.

Operational Ease of Use

Configuring traffic filters can be extremely complex. Consider a NPB that automates the entire traffic filtering process, to eliminate complexity. Do not settle for a solution that requires your team to manually configure and maintain traffic filtering.

No Dropped Packets

Traffic filtering can be computationally difficult if not implemented well in an NPB. Look for NPBs that maintain the complete packet stream when filtering is enabled. An NPB should NEVER drop packets, not even when users are making multiple simultaneous traffic filtering changes to the NPB configuration.

Simultaneous Feature Usage

NPBs are typically used by multiple teams within an IT organization. You will want an NPB that supports the use of multiple features, such as application layer filtering and deduplication, without reducing performance or introducing potential configuration errors. Without the power to deliver the features needed by all your teams, you may not get the results you bargained for.

MORE INFORMATION

Find out more about the advantages of advanced traffic filtering:

- White paper: [Supercharge Your Monitoring Tools](#)
- Solution brief: [Understanding How Feature Combinations Affect NPB Performance](#)

Visibility Architecture

Visibility is defined by Webster as the “capability of being readily noticed” or “the degree of clearness”. For network or application visibility, we are talking about removing blind spots that are hiding the ability to readily see security threats and resolve performance issues.

DEFINITION OF VISIBILITY ARCHITECTURE

A visibility architecture is the end-to-end infrastructure, including hardware and software, which provides access to the data moving through your physical and virtual network devices. While it is possible to deploy visibility components piecemeal as you solve one problem after another, this won't give you a cohesive strategy and can lead to unnecessary complexity and higher operating costs. Instead of just adding components at crisis points, lay a foundation that gives you total access to packet data and metadata and can support any future security tool or solution you may want to deploy. Visibility is created by deploying physical and virtual network taps, bypass switches, and packet processors (NPBs) to access, filter, and process raw data from relevant segments of your network.

A visibility architecture can eliminate blind spots, control costs while maximizing ROI, and simplify operations.

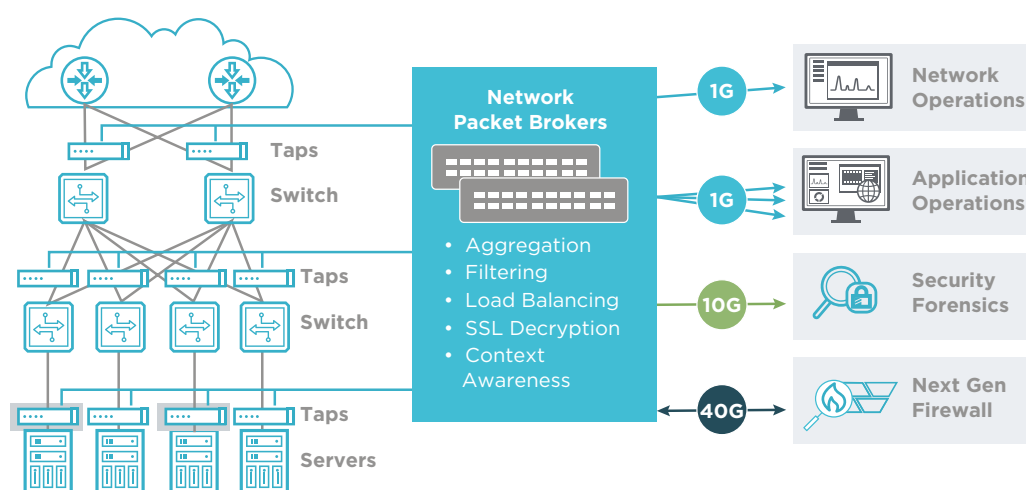


Figure 12. A visibility architecture combines traffic access, packet filtering, and data insight



Figure 12 illustrates the functioning of a visibility architecture. The architecture on the left illustrates data access from across multiple network segments, using taps, virtual taps, and bypass switches.

In the middle is the network packet broker (NPB), capable of data aggregation, customizable filtering, deduplication, and load balancing. These features ensure your security tools get the data they need without being overwhelmed. An important feature of some NPBs is context-awareness or application layer intelligence, which helps you know when something unusual is taking place on your network. Being able to identify traffic by the type of application, endpoint device, bandwidth consumed, geolocation of users, or content of an encrypted packet gives you more information to thwart cyberattacks.

On the right is a sample of the security and monitoring tools that are fed by the visibility platform.

USE CASES AND BENEFITS

Protect and secure the network

To effectively monitor network security and performance, you need access to all the data traversing your network to identify threats and zero in on alerts that need attention. Blind spots where you cannot directly access portions of your network traffic leave you vulnerable to attacks. You need your security appliances and solutions to see all the traffic that impacts your organization. Lack of visibility can lead to network problems and outages, increased risk of cyberattacks, and regulatory compliance issues.

Reduce cost of security monitoring

A visibility platform also helps reduce the cost of inline security monitoring by helping your tools work more efficiently. Once aggregated, the NPB filters and processes your traffic to provide each security tool with only the data it needs, eliminating unnecessary packets. With a condensed workload, your tools operate with less congestion and last longer. NPB load balancing helps you use your tools more efficiently and delay the purchase of additional capacity.

Simplify management of security architecture

Because a visibility platform is deployed between raw data traffic and your security tools, it provides a centralized management and control point. Policies configured at your NPB can be consistently applied across your security tools. Make changes centrally at the NPB and they are broadcast to all your security tools—both inline and out-of-band. When it makes sense to have traffic identified by one tool sent directly to another tool (known as service chaining), the NPB can make it happen automatically.

Enable better forensics analysis

Complete visibility is also critical to out-of-band security forensics and analysis that can determine where new threats are coming from. These type of security tools are more frequently providing input to inline security solutions to accelerate the response to a security attack. Research shows that organizations are still struggling to identify when they've been breached and to stop the attack.

Troubleshoot network performance

In many cases, application performance is tied to network performance. When applications run slowly or stop working, you need real-time network diagnosis to pinpoint the root cause. A proper visibility solution allows you to immediately isolate the application, user, geo-location, and device experiencing problems—so you can identify and resolve the issue promptly.

CONSIDERATIONS WHEN EVALUATING VISIBILITY ARCHITECTURES

When considering visibility architectures, there are several items to investigate. Here is a short list of common items:

Flexibility

You will want, and need, options for evolving your visibility platform over time. This includes the flexibility to deploy inline and out-of-band visibility solutions and the ability to monitor both physical and virtual traffic. Application Intelligence is another area to look for. While you may not want to engage in all of these activities right away, you should look for a solution that allows you to add the pieces you want, when you want, without a forklift upgrade.

Ease of Use

This will be a critical component that will heavily influence your total cost of ownership (TCO). Look for a solution that uses a graphical, drag and drop interface. A command line interface will take you 10 times (or more) longer than a drag and drop interface to configure filters. The management system should also be able to handle everything—from global element management, to policy and configuration management, to data center automation and orchestration management. Engineering flexible management for network components will be a determining factor in how well your network scales.

Technology

A third consideration is around the technology. While Vendor products may sound the same, they usually aren't. In general, look for NPBs that can process traffic at line rate under all conditions. Only a very few NPBs do this. Anything less adds delay to your monitoring effort and reduces network responsiveness for users.

You will also want failover technology that is as fast as possible for inline solutions. Visibility should be supported with external bypass switches for maximum flexibility. This will maximize your fail-over capabilities and minimize loss of data on your network.

MORE INFORMATION ON VISIBILITY ARCHITECTURES

Find out more about the benefits of a visibility architecture for network security:

- White paper: [Inspecting SSL Traffic](#)
- Solution brief: [Security in an Encrypted World](#)
- Solution brief: [Role of Active SSL Decryption in Network Monitoring](#)

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